

L1 – Trig Review and Special Angles

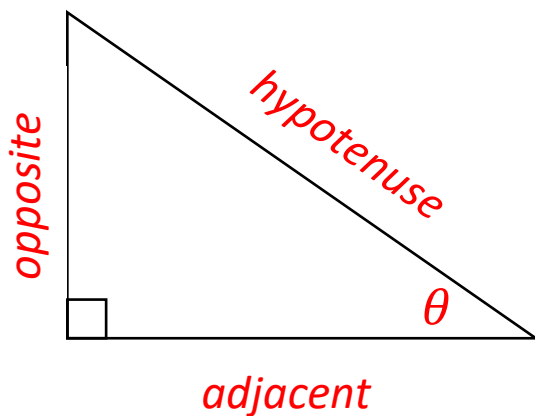
MCR3U

Part 1: Trig Review

Your main takeaway from grade 10 trigonometry should have been:

If we know a right triangle has an angle of θ , all other right triangles with an angle of θ are _____ and therefore have _____ ratios of corresponding sides.

There are three primary trigonometric ratios for right angled triangles. _____, _____, and _____.



$$\sin \theta =$$

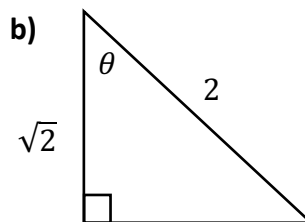
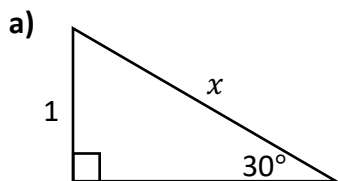
$$\cos \theta =$$

$$\tan \theta =$$

Acronym: SOHCAHTOA

S $\frac{O}{H}$ **C** $\frac{A}{H}$ **T** $\frac{O}{A}$

Example 1: Find the indicated missing side or angle of each triangle



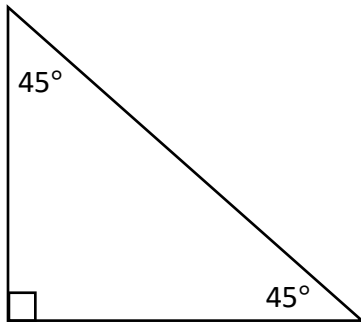
Part 2: Special Angles

There are 2 special triangles:

i) isosceles: $45^\circ - 45^\circ - 90^\circ$

ii) half equilateral: $30^\circ - 60^\circ - 90^\circ$

i)

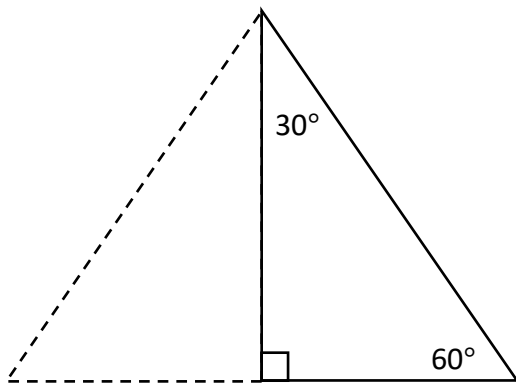


$$\sin 45^\circ =$$

$$\cos 45^\circ =$$

$$\tan 45^\circ =$$

ii)



$$\sin 30^\circ =$$

$$\cos 30^\circ =$$

$$\tan 30^\circ =$$

$$\sin 60^\circ =$$

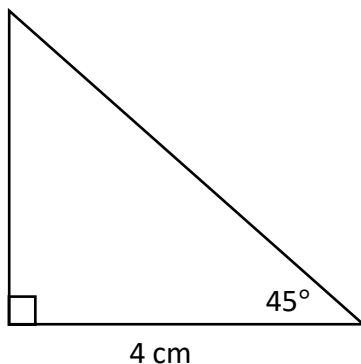
$$\cos 60^\circ =$$

$$\tan 60^\circ =$$

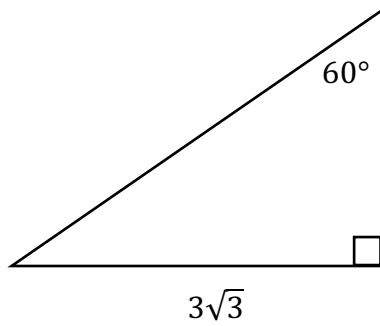
All sized right triangles with these angles are SIMILAR and therefore will have the same ratios of corresponding sides. Therefore, we can use these 2 special triangles to get _____ values for trig ratios involving a 30° , 45° , or 60° reference angle AND we don't need a calculator!

Example 2: Use special triangles to find the EXACT values of all sides and angles

a)



b)



Example 3: Determine the exact value of...

a) $(\sin 45^\circ)(\cos 45^\circ) + (\sin 30^\circ)(\sin 60^\circ)$

b) $\frac{\sin^2 30^\circ}{1 - \cos 30^\circ}$

Part 3: Rationalizing the Denominator

Fractions should be simplified so that the denominator contains only rational numbers.

Example 4: Rationalize the denominator for each of the following expressions

a) $\frac{1}{\sqrt{2}}$

b) $\frac{3}{1 + \sqrt{5}}$